

RAHUL KUMAR

Aspiring Data Scientist | Machine Learning & MLOps

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SUMMARY

Computer Science graduate (B.Tech) skilled in Python, SQL, and Scikit-learn, with Data Science internship experience in customer churn prediction. Built 3 end-to-end ML pipelines with MLOps practices (DVC, Docker, CI/CD), from raw data to deployment. Seeking a Machine Learning Engineer or entry-level Data Science role to deliver measurable business impact.

TECHNICAL SKILLS

- **Programming & Querying:** Python (Pandas, NumPy), SQL (JOINS, Aggregations), HTML5
- **Data Analysis & Visualization:** Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Statistical Analysis
- **Machine Learning:** Scikit-learn, XGBoost, Predictive Modeling, Regression, Classification, Decision Trees, Feature Engineering, Cross-Validation, Model Deployment
- **Tools & Deployment:** Git, GitHub, Docker, DVC, Flask, FastAPI, GitHub Actions (CI/CD), SQL Server, VS Code
- **Core Competencies:** Data Structures & Algorithms (DSA), OOPs, Data Cleaning, Model Evaluation
- **Currently Learning:** Deep Learning (Neural Networks)

EXPERIENCE

Data Science Intern — Renu Sharma Healthcare and Educational Foundation Aug 2026 – Sep 2026

- Developed an end-to-end customer churn prediction project: data cleaning, EDA, feature engineering, and model evaluation in Python.
- Built and compared classification models (Scikit-learn, XGBoost) to identify at-risk customers, tracking ROC-AUC, precision, recall, and F1-score.

Industrial Trainee — Indian Oil Corporation Limited (IOCL), Barauni Refinery Jun 2025 – Jul 2025

- Analyzed refinery system data for the Information Systems team, identifying inconsistencies that improved reporting accuracy.
- Structured raw operational data into a technical report used by senior management for data-driven decisions.
- Built HTML5 frontend components to improve usability of an internal refinery application for non-technical staff.

PROJECTS

Customer Churn Prediction Pipeline (ChurnGuard)

Python, XGBoost, Pandas, DVC, Flask, Docker | github.com/Rahul117590/churnguard-ml

- Analyzed ~7,000 telecom customer records and benchmarked 4 classification models; selected XGBoost for the best ROC-AUC and F1-score on the churned class.
- Engineered features (One-Hot Encoding, StandardScaler) in a modular, DVC-versioned pipeline (ingestion, preprocessing, training, evaluation) for fully reproducible experiments.
- Deployed the model as a Dockerized Flask web app so telecom teams can flag at-risk customers and act before they churn.

Loan Approval Prediction

Python, Scikit-learn, Random Forest, DVC | github.com/Rahul117590/Loan-Approval-Prediction-using-Decision-Tree

- Built a Random Forest classification pipeline to automate loan Approve/Reject decisions from applicant and asset data, mirroring NBFC underwriting workflows.
- Achieved 98.01% accuracy, 0.976 precision, and 0.993 recall, reducing manual review effort and enabling faster, consistent decisions.
- Organized the code into a DVC-managed src/ pipeline with console and file logging for full traceability.

Breast Cancer Prediction (Malignant vs Benign Classification)

Python, Scikit-learn, AdaBoost, DVC, Flask | github.com/Rahul117590/breast-cancer-prediction-adaboost

- Built an AdaBoost ensemble classifier on the Breast Cancer Wisconsin dataset to distinguish malignant from benign tumors, using a modular src/ pipeline (ingestion, preprocessing, feature engineering, training, evaluation).
- Tuned the model with feature scaling and cross-validation; added structured logging and DVC-versioned stages for reproducible experiments.
- Deployed via a Flask web interface for quick predictions from patient feature inputs, supporting early diagnosis.

EDUCATION

B.Tech, Computer Science & Engineering, BBDIT, Ghaziabad, UP — CGPA: 7.04/10 2022 – 2026

Higher Secondary (10+2), BR DAV Public School, Begusarai, Bihar — 77.6% 2022

Secondary (10th), Nezamia Public School, Patna, Bihar — 83.4% 2020

ACHIEVEMENTS

- Practicing Data Structures & Algorithms (DSA) consistently to strengthen programming fundamentals for technical interviews.
- Won chess competitions at both school and college level, reflecting strategic thinking and problem-solving that carries into structuring and debugging ML pipelines.

CERTIFICATIONS

- SQL Fundamentals | Machine Learning Fundamentals | Pandas | Deep Learning Fundamentals — Kaggle Learn